

MAT 535 HW05

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Problem 1 (§13.5#5). For any prime p and any nonzero $a \in \mathbb{F}_p$ prove that $x^p - x + a$ is irreducible and separable over $\mathbb{F}_p$.

Solution. Let r be a root of $f(x) = x^p - x + a$ in some extension of $\mathbb{F}_p$ (noting that $r \notin \mathbb{F}_p$ because elements of $\mathbb{F}_p$ makes $x^p - x$ zero), and consider the extension $\mathbb{F}_p(r)$. Observe that for any $k \in \mathbb{F}_p$,

$$f(r+k) = (r+k)^p - (r+k) + a = r^p - r + a + k^p - k = 0.$$

If $m_r(x)$ is the minimal polynomial of r over $\mathbb{F}_p$, then $m_{r+k}(x) = m_r(x-k)$ for $k \in \mathbb{F}_p$, and $m_{r+k}(x)$ divides $f(x)$. For a given $k \in \mathbb{F}_p$ let $S_k \subseteq \mathbb{F}_p(r)$ be the roots of $m_{r+k}(x)$; any two sets S_k are either disjoint or equal, by the minimality of the minimal polynomial, and the S_k are all equal in size because the $m_{r+k}(x)$ all have the same degree. Thus, after deleting repeats, the sets $S_0, \dots, S_{p-1}$ partition $\{r, r+1, \dots, r+p-1\}$ into sets of equal size. This means $|S_0| = \deg m_r(x)$ divides p , so it is either p or 1 , but we know that $r \notin \mathbb{F}_p$, so it must be p , i.e. $m_r(x) = f(x)$.

The work above also shows that $f(x)$ is separable because its roots $r, r+1, \dots, r+p-1$ are pairwise distinct. □

Problem 2 (§13.5#7). Suppose K is a field of characteristic p which is not a perfect field, i.e. $K \neq K^p$. Prove that there exist irreducible inseparable polynomials over K . Conclude that there exist inseparable finite extensions of K .

Solution. Let $a \in K - K^p$ and consider the polynomial $f(x) = x^p - a$. If r is a root of f in some extension L/K , then $f(x)$ factors over L as $f(x) = (x-r)^p$. This tells us that f does not factor over K because any such factor would be of the form $(x-r)^k$ for $0 < k < p$ and the coefficient of x^{k-1} , which is $-kr$, is not in K because r isn't in K . Thus f is irreducible and its factorization over L has multiple roots, so it (and the extension L) is inseparable. □

Problem 3 (§14.1#4). Prove that $\mathbb{Q}(\sqrt{2})$ and $\mathbb{Q}(\sqrt{3})$ are not isomorphic.

Solution. Any homomorphism $\varphi: \mathbb{Q}(\sqrt{2}) \rightarrow \mathbb{Q}(\sqrt{3})$ needs to send 1 to 1 , hence fixes $\mathbb{Q}$. Therefore the only leeway is in the image of $\sqrt{2}$, which satisfies $\varphi(\sqrt{2})^2 = \varphi(2) = 2$. However, 2 is not a square in $\mathbb{Q}(\sqrt{3})$, since for rational a, b ,

$$(a + b\sqrt{3})^2 = a^2 + 3b^2 + 2ab\sqrt{3} \in \mathbb{Q}$$

if and only if at least one of a, b is zero, hence either $2 = a^2$ or $3b^2$, both of which we know don't have rational solutions from the standard elementary number theory argument. □

Problem 4 (§14.1#5). Determine the automorphisms of the extension $\mathbb{Q}(\sqrt[4]{2})/\mathbb{Q}(\sqrt{2})$ explicitly.

Solution. Any automorphism $\varphi: \mathbb{Q}(\sqrt[4]{2}) \rightarrow \mathbb{Q}(\sqrt[4]{2})$ fixes the prime subfield $\mathbb{Q}$. In order to also fix $\sqrt{2}$ we must have $\varphi(\sqrt[4]{2})^2 = \varphi(\sqrt{2}) = \sqrt{2}$, so $\sqrt[4]{2}$ maps to one of the two solutions to $x^2 = \sqrt{2}$, i.e. either $\varphi(\sqrt[4]{2}) = \sqrt[4]{2}$ or $-\sqrt[4]{2}$. Since $\mathbb{Q}(\sqrt[4]{2})$ is generated as a $\mathbb{Q}(\sqrt{2})$ -algebra by $\mathbb{Q}$ and $\sqrt[4]{2}$, this choice of $\varphi(\sqrt[4]{2})$ uniquely determines the automorphism (and we can check manually that both choices of $\varphi(\sqrt[4]{2})$ do produce automorphisms fixing $\mathbb{Q}(\sqrt{2})$). In other words, $\text{Aut}(\mathbb{Q}(\sqrt[4]{2})/\mathbb{Q}(\sqrt{2}))$ consists of the two automorphisms

$$\begin{aligned}\varphi_1(a + b\sqrt[4]{2}) &= a + b\sqrt[4]{2} \\ \varphi_2(a + b\sqrt[4]{2}) &= a - b\sqrt[4]{2}\end{aligned}$$

□

where $a, b \in \mathbb{Q}(\sqrt{2})$.

Problem 5 (§14.1#8). Prove that the automorphisms of the rational fraction field $k(t)$ which fix k are precisely the **fractional linear transformations** determined by $t \mapsto \frac{at+b}{ct+d}$ for $a, b, c, d \in k$ and $ad - bc \neq 0$.

Solution. Let $\varphi: k(t) \rightarrow k(t)$ be an automorphism and $\varphi(t) = \frac{p(t)}{q(t)}$ for relatively prime polynomials $p(t), q(t) \in k[t]$. We can see that $k(t)$ is an extension of $k(y)$ where $y = \varphi(t)$, and this extension is algebraic because t is a root of the polynomial

$$F(x) = y \cdot q(x) - p(x) \in k(y)[x].$$

To show that F is irreducible over $k(y)$, it suffices by Gauss's lemma to prove that it is irreducible over $k[y]$. Then, if $F(x) = a(x)b(x)$ for $a(x), b(x) \in k[y][x]$, one of $a(x), b(x)$ has degree 1 as a polynomial in y over $k[x]$ and the other has y -degree 0, i.e. $F(x) = a(x) \cdot (c(x)y + d(x))$ for some $c(x), d(x) \in k[x]$. This then implies that $a(x) \in k$ because $p(x)$ and $q(x)$ are coprime, so $F(x)$ is irreducible over $k[y]$.

Observe now that the degree of $F(x)$ is $\max(\deg p, \deg q)$ because no two terms in the subtraction $yq(x) - p(x)$ can cancel (since the terms with a y in them can't cancel with the terms with no y), so

$$\max(\deg p, \deg q) = [k(t) : k(\varphi(t))].$$

On the other hand $k(t)$ should be equal to $k(\varphi(t))$ because φ is an automorphism, so $[k(t) : k(\varphi(t))] = 1$, hence $\varphi(t) = \frac{at+b}{ct+d}$ for some $a, b, c, d \in k$ with c and d not both zero. If $ad - bc = 0$, then if $c \neq 0$ we have

$$\frac{at+b}{ct+d} = \frac{at+bc/c}{ct+d} = \frac{at+ad/c}{ct+d} = \frac{a(ct+d)}{c(ct+d)} = \frac{a}{c}$$

which means that φ maps $k(t)$ into k , so such a, b, c, d make φ not an automorphism.

On the other hand if $ad - bc \neq 0$ then we can define an inverse to φ by sending t to

$$\psi(t) = \frac{dt - b}{-ct + a}.$$

This is an inverse because

$$\begin{aligned} \psi(\varphi(t)) &= \psi\left(\frac{at+b}{ct+d}\right) = \frac{d(at+b) - b(ct+d)}{-c(at+b) + a(ct+d)} = \frac{(ad-bc)t}{ad-bc} = t, \\ \varphi(\psi(t)) &= \varphi\left(\frac{dt-b}{-ct+a}\right) = \frac{a(dt-b) + b(-ct+a)}{c(dt-b) + d(-ct+a)} = \frac{(ad-bc)t}{ad-bc} = t. \end{aligned}$$

Thus the fractional linear transformations are the automorphisms of $k(t)$. □